PROJECT REPORT

EV MARKET INTELLIGENCE DASHBOARD

Data Visualization and Business Intelligence using Tableau Public

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Team ID: [INSERT ACTUAL TEAM ID]

Dataset: ev_market_test.csv
Dashboard URL: [INSERT VERIFIED TABLEAU PUBLIC URL]
GitHub repository: [INSERT REPOSITORY URL]
Project demo video: [INSERT PUBLIC VIDEO URL]

1. INTRODUCTION

1.1 Project Overview

The EV Market Intelligence Dashboard is a Tableau Public data-visualization project built from a supplied electric-vehicle dataset. It presents market and vehicle attributes through nine visualisations covering category sales, manufacturer performance, battery technology, driving range, manufacturing geography, customer ratings, sales by model year, battery capacity versus range, and reported CO_2 savings.

The supplied CSV contains 100,000 records and 37 fields, with a file size of 23,238,435 bytes (22.16 MiB). Fields include manufacturer, model year, manufacturing country, category, battery capacity/type, range, annual sales units, customer rating and CO_2 saved per year.

1.2 Purpose

The purpose is to turn a large flat dataset into an accessible decision-support dashboard for three stakeholder scenarios: an EV market strategy director, an automotive business intelligence manager, and a sustainable mobility research analyst. The dashboard helps users compare segments and identify patterns. It is descriptive and exploratory; it does not prove causation or independently validate the dataset's reported values.

2. IDEATION PHASE

2.1 Problem Statement

Stakeholders need a consolidated view of EV market performance across vehicle categories, manufacturers, battery technologies, geography, range, customer satisfaction and sustainability metrics. Raw tabular data makes cross-segment comparisons slow and difficult. The project addresses this by building a single dashboard containing nine complementary visualisations, enabling users to investigate the questions relevant to their roles.

Stakeholder scenarios

Scenario / persona	Main questions	Dashboard support
Emily Johnson — Market Strategy Director	Which categories and manufacturers lead? How do range, battery adoption and satisfaction vary?	Sales rankings, battery mix, range by category, ratings and model-year trend.
David Chen — Automotive BI Manager	What is the competitive landscape? Which manufacturers and countries have stronger reported sales?	Top-10 manufacturer chart, category sales, country map, battery mix and model-year trend.
Dr. Sarah Williams — Sustainable Mobility Research Analyst	How do technology, range, manufacturing geography and reported CO_2 savings vary?	Battery mix, capacity-versus-range scatter, range by category, country map and CO_2 savings chart.

2.2 Empathy Map Canvas

Stakeholder lens	What the stakeholder may think / need	What the dashboard provides
Says	"Show me which segments and manufacturers lead."	Ranked sales bars and top-10 manufacturer view.
Thinks	"Are the differences meaningful, and what does the data actually measure?"	Clear aggregation labels, units, tooltips and explicit caveats.
Does	Filters categories, compares manufacturers, inspects outliers and checks geography.	Interactive worksheets and dashboard filters where configured.
Feels	Needs confidence that the numbers are understandable and not misleading.	Consistent units, transparent measure definitions and restrained claims.

This empathy map is a design aid derived from the three supplied scenario descriptions; it is not based on interviews with real stakeholders.

2.3 Brainstorming

- Use sorted horizontal bars for category and manufacturer comparisons.
- Use a pie chart to show the count of records by battery type; do not label it market share by sales.
- Use a bar chart for average range and average customer rating by category.
- Use a geographic map for manufacturing-country distribution, with the legend stating the selected measure.
- Use a line chart to show annual sales totals by model year.
- Use a scatter plot to inspect the relationship between battery capacity and driving range.
- Use a bar chart for average reported CO_2 savings by category.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Step	User activity	Pain point	Dashboard response
1. Define question	Select a market, competitive or sustainability question.	Raw data has many fields and rows.	Start with role-based questions and clear chart titles.
2. Find relevant metric	Identify sales, range, ratings, battery or CO_2 field.	Measures can be confused with counts or averages.	Use explicit axis titles, units and aggregation labels.
3. Compare segments	Compare categories, manufacturers or countries.	Sorting and cross-comparison are slow in tables.	Ranked bars, map and category comparisons.
4. Investigate detail	Inspect a category, model or outlier.	Aggregate charts can hide individual records.	Use tooltips and filters where configured.
5. Make a decision	Form a product, market or research hypothesis.	Dataset may not represent validated real-world market data.	Use insights as exploratory evidence and validate before investment/policy decisions.

3.2 Solution Requirement

Type	Requirement
Functional	Load the supplied CSV in Tableau Public.
Functional	Create nine visualisations addressing the three scenarios.
Functional	Show sales, manufacturer, battery, range, geography, ratings, model-year, capacity/range and CO_2 measures.
Functional	Provide readable titles, axis labels, legends and useful tooltips.
Functional	Combine the worksheets into a dashboard and publish it to Tableau Public.
Non-functional	Dashboard should be legible at a selected desktop size without overlapping or clipped content.
Non-functional	Use consistent number formats, colour meanings and metric definitions.
Evidence	Attach screenshots and actual URLs for the published dashboard, repository and demo video.

3.3 Data Flow Diagram

Source CSV → Tableau Public data connection → field/type and data-quality checks → worksheets (nine visualisations) → dashboard layout and interactions → Tableau Public published view → stakeholders inspect and interpret results.

The current project flow does not include a Snowflake warehouse, SQL pipeline, API, or automated ETL process. Those should not be described as implemented components.

3.4 Technology Stack

Component	Use in project
CSV dataset	Source data: ev_market_test.csv.
Tableau Public (web authoring)	Data connection, worksheet authoring, dashboard assembly and publishing.
Web browser	Access Tableau Public and published dashboard.
Microsoft Word / PDF	Project documentation and report export.

GitHub (planned / to be confirmed)	Versioned project artefacts, if repository has been created.
Screen recording tool (planned / to be confirmed)	Narrated walkthrough of the dashboard, if recorded.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The dashboard fits the problem because it reduces the need to inspect multiple raw-data tables and puts the requested measures into a single visual interface. Each chart answers a distinct question. It supports exploratory comparison rather than forecasting or causal analysis.

4.2 Proposed Solution

No.	Visualisation	Build logic / question answered
1	Annual sales by category	Horizontal bars: Category vs SUM(Annual Sales Units), sorted descending.
2	Top 10 EV manufacturers	Horizontal bars: Manufacturer vs SUM(Annual Sales Units), Top 10 filter, descending sort.
3	Battery technology mix	Pie: Battery Type as colour, COUNT(Vehicle Id) as angle/size; record mix, not sales share.
4	Driving range by category	Bars: Category vs AVG(Range Km), sorted descending.
5	Manufacturing country map	Map: Manufacturing Country; size/colour based on a clearly stated measure such as SUM(Annual Sales Units).
6	Customer satisfaction	Bars: Category vs AVG(Customer Rating); note that ratings are tightly clustered.
7	Sales by model year	Line: Model Year vs SUM(Annual Sales Units).
8	Battery capacity vs range	Scatter: Battery Capacity Kwh vs Range Km, coloured by Battery Type.
9	Sustainability impact	Bars: Category vs AVG(CO2 Saved Tons Per Year), descending.

4.3 Solution Architecture

The architecture is intentionally lightweight: CSV file → Tableau Public data source → worksheet-level aggregations and visual encodings → tiled dashboard → published Tableau Public URL. There is no separate server-side data model or database in the implemented path described here.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Tasks	Completion evidence
1. Understand problem	Review overview and three stakeholder scenarios.	Problem statement and question list.
2. Inspect data	Load CSV; inspect field names, types, nulls and duplicates.	Data source screenshot and quality summary.
3. Build worksheets	Create and validate nine visualisations.	Nine named worksheets.
4. Assemble dashboard	Choose fixed desktop size, use tiled layout, set Fit options and remove overlap.	Full dashboard screenshot.
5. Validate	Check sort order, aggregation, units, legends, tooltips, filters and clipping.	Testing checklist and screenshots.
6. Publish	Publish dashboard to Tableau Public	Verified public URL.

	and test the public URL.	
7. Document and submit	Complete templates, add evidence, repository and video links.	Final report, GitHub repository and demo video.

This is a practical work breakdown, not a verified historical time log. Replace or annotate it if your course requires actual dates and durations.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Data-quality checks were run against the supplied CSV file. They do not represent a Snowflake performance test.

Test	Observed result / acceptance criterion
Data rendered	100,000 rows, 37 fields; file size 23,238,435 bytes (22.16 MiB).
Missing values	9,511 missing cells across 5 columns. Handle per metric; do not replace all nulls with zero.
Duplicate rows	0 full-row duplicates found.
Duplicate vehicle IDs	0 duplicate vehicle_id values found.
Aggregation correctness	Verify sales charts use SUM(Annual Sales Units); range/ratings/CO_2 charts use AVG of the corresponding measure; battery mix uses COUNT(Vehicle Id).
Filter behaviour	For every filter actually present, verify only intended worksheets change. Record timings only if measured.
Dashboard rendering	At presentation size, all nine views, legends and labels should be visible without overlap or unintended scrolling.
Query history	Not applicable to CSV-based Tableau Public workflow; no Snowflake query history or warehouse execution time is available.

No custom calculated fields are confirmed by the available project evidence. The chart definitions above mainly use Tableau's built-in aggregations. If custom fields were added, list their exact names and formulas and capture a screenshot.

7. RESULTS

The following figures are computed from the supplied CSV. They describe the dataset, not independently verified real-world EV market performance.

Finding	Dataset result	Interpretation
Highest category sales	Two-Wheeler Scooter — 85,031,983 units	Largest total in the provided data.
Lowest category sales	Electric Bus — 5,323,646 units	Smallest total in the provided data.
Top manufacturer	SurgeWorks — 4,883,639 units	Highest manufacturer total in the top-10 ranking.
Most common battery type	NMC — 33,836 records	Record count, not market share by sales.
Highest average range	Luxury Sedan — 549.2 km	Category average from available range values.
Lowest average range	Two-Wheeler Scooter — 105.0 km	Category average from available range values.
Highest average customer rating	Crossover — 4.095	Differences across categories are small.
Highest annual sales by model year	2021 — 88,088,210 units	Year with the highest summed annual sales in this dataset.
Highest average reported CO_2 savings	Electric Bus — 9.05 tons/year	Dataset-reported average; not independently verified environmental impact.
Highest country sales total	China — 78,870,003 units	Based on the manufacturing-country field in the dataset.

7.1 Output Screenshots

[INSERT SCREENSHOT 1: full nine-visualisation dashboard]

[INSERT SCREENSHOT 2: published Tableau Public dashboard in view mode]

[INSERT SCREENSHOT 3: data source / record count and data-quality evidence]

[INSERT SCREENSHOT 4: selected filters working, if filters are present]

Screenshots were not included with the Final Report Template upload, so these evidence slots are intentionally left for the actual project screenshots.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Combines nine relevant views in one place and reduces manual comparison of raw rows.
- Sorted bars make rankings and category differences easy to scan.
- Scatter plot helps identify range/capacity patterns and outliers.
- Map adds a geographic perspective to manufacturer distribution.
- Tableau Public enables sharing a published interactive dashboard without requiring recipients to install Tableau Desktop.

Disadvantages / Limitations

- Results depend on the quality and representativeness of the supplied dataset; market totals should not be treated as verified industry statistics.
- Battery-type pie uses record counts unless deliberately changed to a sales measure; it is not automatically market share.
- Customer ratings are tightly clustered, so ranking small differences may overstate practical significance.
- A scatter plot shows association, not causation.
- Tableau Public web authoring and browser rendering can be slow with dense marks; labels, legends and dashboard sizing need manual adjustment.
- CSV-based implementation has no Snowflake query history, warehouse monitoring, or database-side performance testing.
- Public publishing means the dashboard URL is shareable; do not publish confidential data.

9. CONCLUSION

The EV Market Intelligence Dashboard demonstrates how a supplied 100,000-record, 37-field CSV can be transformed into nine complementary Tableau visualisations. The charts address the three supplied scenarios by covering sales, competition, battery adoption, range, manufacturing geography, customer ratings, model-year trends and reported CO_2 savings. The main deliverable is an exploratory dashboard; the findings should be validated against reliable external sources before being used for commercial, policy or investment decisions.

10. FUTURE SCOPE

- Add carefully scoped dashboard filters and actions, then test their impact on all relevant views.
- Add KPI cards for total annual sales, number of manufacturers, average range and average customer rating, with precise definitions.
- Compare battery technologies on range, capacity, cycle life, degradation and charging metrics rather than record count alone.
- Use verified external datasets to validate country totals, sales definitions and sustainability measures.
- Add time-series forecasting only if the dataset has appropriate time coverage and a validated modelling method.

- Improve dashboard responsiveness and test the published view at laptop and mobile widths.
- Create a Tableau Story or narrated screen recording to explain how each visual answers a stakeholder question.

11. APPENDIX

Source Code (if any)

No separate application source code is required for the current Tableau-only implementation. The Tableau workbook is the primary visualisation artefact. Insert the downloaded .twb file in the repository if available.

Dataset Link

Dataset file: ev_market_test.csv (100,000 records, 37 fields). Repository or cloud link: [INSERT DATASET LINK OR GITHUB PATH].

GitHub & Project Demo Link

GitHub repository: [INSERT PUBLIC REPOSITORY URL]

Tableau Public dashboard: [INSERT VERIFIED DASHBOARD URL]

Project demo video: [INSERT PUBLIC GOOGLE DRIVE / VIDEO URL]

Team ID: [INSERT ACTUAL TEAM ID]

Metric Definitions

Metric	Definition used
Annual sales	SUM(annual_sales_units) unless otherwise stated.
Battery technology mix	COUNT(vehicle_id) grouped by battery_type.
Average driving range	AVG(range_km) grouped by category.
Average customer rating	AVG(customer_rating) grouped by category.
CO_2 savings	AVG(co2_saved_tons_per_year) grouped by category.

Report prepared from the supplied report outline and ev_market_test.csv. Placeholder fields must be completed with real project evidence before submission.